IKT215 Exercise 6: Non-parametric Classification

February 26th, 2026

Exercise sessions, while not mandatory, are an integral part of your learning experience and exam preparation. Some exam questions will be based on tasks covered during these sessions. Although you have the flexibility to complete exercises at home and ask questions later (preferably within one week), I strongly recommend attending the sessions in person for immediate support and guidance. This allows you to benefit from direct interaction and real-time problem-solving. I respond promptly to emails (manuel.s.mathew@uia.no).

Question 1: Cubic Histogram Classification Rule

Dataset: banknote_authentication.csv
Description: banknote_authentication.names

1. Select two variance features (Variance, Skewness)
2. Implement and visualize cubic histogram classification with:
    a. Equal-width binning (3 bins/axis → 9 cells),
    b. Max-count class assignment per cell.
3. Test accuracy with different bin configurations:
    a. 2×2 bins → 4 cells
    b. 4×4 bins → 16 cells
4. Analyze:
    a. how bin granularity affects overfitting?
    b. why this method is impractical for high-dimensional data?

Question 2: kNN Classification

Dataset: iris.csv
Description: iris.names

1. Split the data in the ratio 4:1
2. Implement and visualize kNN with an appropriate k value and Euclidean distance.
3. Report:
    a. Confusion matrix
    b. Class-wise precision/recall
4. Analyze:
    a. which classes are most confused and why?
    b. how increasing k to 15 changes the results?
5. Vary the distance measure and investigate how this changes the results.

Question 3: Weighted kNN Classification with distances

Dataset: abalone.csv (Physical measurements → Age ≥10 rings)
Description: abalone.names

1. Preprocess: Normalize features
2. Compare performance using:
    a. Manhattan vs. Cosine distance
    b. Uniform vs. inverse-distance weighting
3. Analysis:
    a. Which weighting scheme handles imbalanced age groups better?
    b. Why does Cosine distance outperform Manhattan here? (Hint: directionality)

Question 4: Kernel Classification

Dataset: Synthetic Data

1. Generate synthetic data
2. Implement different kernel classifiers
3. Calculate accuracy
4. Visualize the results
5. Test with different kernel classifiers